
Decision on your submission to Scientific Reports

1 message

Scientific Reports <srep@nature.com>

Fri, Jul 3, 2026 at 8:57 AM

To: reza_abiri@uri.edu

Ref: Submission ID 3e91b1d6-53d0-4395-a9bd-ecd0a9e4b8c3

Dear Dr Abiri,

Your manuscript "Sensor-based Bruxism Detection with Dual-Branch WaveletCNNs and Audio-EMG Data Fusion" has now been assessed. If there are any reviewer comments on your manuscript, you can find them at the end of this email.

Regrettably, your manuscript has been rejected for publication in Scientific Reports.

Editor's Comments

"Recommendation: Reject"

- María Pilar Pecci Lloret

Thank you for the opportunity to review your work. I'm sorry that we cannot be more positive on this occasion and hope you will not be deterred from submitting future work to Scientific Reports.

Kind regards,

Shruti Anand
Assistant Editor
Scientific Reports

Reviewer Comments:

Reviewer 1

Good work done, I compliment all authors for the innovative thinking put into this study.

Reviewer 2

Despite interesting, this manuscript has many pitfalls that denote some inaccuracies as far as the current standard of care on bruxism knowledge is concerned.

- First of all, the authors seem to rely on the old concept of bruxism=tooth contact. This is not in line with current standard of care and terminology, which include bracing and thrusting (i.e, activities without teeth contact) . Based on that, the authors should consider their paper as an investigation on teeth contact bruxism activities. The title and text should be modified consequently.

- Bruxism is a very vague definition and inclusion criterion in your investigation. Which phenotype are you referring to? Grinding? Bracing? Sleep? Awake? There is a strong push for a better specification in the recent literature. For instance, bruxism is currently considered the medical gateway for dentistry (see recent editorial Cranio. 2024 May;42(3):251-252. doi: 10.1080/08869634.2024.2320624 and many others). Please discuss the rationale of your study in the light of this perspective

- Please refer to the biopsychosocial model of bruxism for any discussion

- Please refer to the INfORM/IADR key points for standard of care TMD management when discussing the possible "TMD" consequences of bruxism

- There are many recent papers discussing the need to improve knowledge in the bruxism field and use appropriate educational and terminological strategies (Mungia et al, CRANIO 2025; Nasanen et al, 2025; Sangalli et al, Cranio 2025) Please give a look at them and discuss

- Several papers by Colonna and Bracci have suggested measurement strategies and observed natural fluctuation in the home environment that can also be applied to sleep studies. Please discuss

- The validity of the device under testing should be reappraised/discussed based on the above concerns

Reviewer 3

Accepted with minor changes

Attachments:

- [https://urldefense.com/v3/__https://reviewer-feedback.springernature.com/download/attachment/4ffeae03-6040-45c0-aa84-de911a0b0bc6__;!!Jh1S!k-bc_1nJFPVkvzlrEJ_hKJJ7DzhFM7kHY-qZdeX_vsZRAMxVVy1UIQX0I11Ft_hfOQ2_9XrdXUN5\\$](https://urldefense.com/v3/__https://reviewer-feedback.springernature.com/download/attachment/4ffeae03-6040-45c0-aa84-de911a0b0bc6__;!!Jh1S!k-bc_1nJFPVkvzlrEJ_hKJJ7DzhFM7kHY-qZdeX_vsZRAMxVVy1UIQX0I11Ft_hfOQ2_9XrdXUN5$)

Reviewer 4

The manuscript is a pilot study presenting a method for discriminating among different muscular activities. While the method is interesting, several factors significantly reduce the interpretability of the results and the study's relevance.

Overall, the introduction and the method are very well presented and very clear. The study aims to propose a convenient method for detecting bruxism, particularly at night. One study reports over 90% accuracy in differentiating 4 movements: jaw opening, jaw clenching, rhythmic teeth grinding, and fatigue/pain.

Comment 1:

The choice of the different classes is important. The authors used different classes than those cited in the literature, including chewing. While chewing is an important function of the jaw, it can be easily assessed by obtaining a patient report or by asking the patient to deactivate the device during eating, since it is a conscious, time-limited activity. It appears that the chewing class inflates the global precision, recall, and F1 reported in this study. Adding a movement that is relatively easy to detect but less relevant can be seen as a way to artificially improve the results of the method presented here.

The authors are also critical of themselves about the differentiation between grinding and clenching, which appears clinically irrelevant when reading the limitation section. In that case, why did they differentiate?

Overall, the authors should discuss the choice of the different classes they included in comparison with the literature. Other classes make it more difficult to compare the results with other studies.

Comment 2:

The authors claim an accuracy of 85% and an F1-score of 80.3% for the detection of bruxism, but those scores are for the classification of events, rather than for the detection of bruxism events vs other events (e.g., movements) and non-events. It is damaging to the study that the "resting" class was not included.

Comment 3:

It would be interesting to compare those results with the study previously cited in the introduction, which achieved 90% accuracy with EMGs alone.

Comment 4:

In the discussion, right after the limitations, rightly stating that the low number of participants does not allow for generalization, the sentence "The leave-one-subject-out validation provides evidence that the learned representations generalize to unseen individuals, supporting the validity of our approach for awake bruxism detection. » is a perfect example of over-generalization. Factors such as age affect EMG accuracy, highlighting inter-individual variability that the current sample (N=5; mean age 38 ± 10 years) is not able to assess. Overall, claims should be reduced to align with the cited limitations.

Comment 5:

Another limitation of this study is that while real patients have been recruited, the bruxism events are only emulated. They are neither data nor discussion of whether the emulated events are similar to natural ones or whether they are potentially overplayed, making their detection easier.